

Ishwariya Venkatesh

Senior Scientist / Scientist D

Council for Scientific and Industrial Research — Center for Cellular and Molecular Biology

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PROFESSIONAL APPOINTMENTS

Senior Scientist (Scientist D) — Center for Cellular and Molecular Biology, CSIR	Dec 2022 – Present
Ramanujan Fellow — Center for Cellular and Molecular Biology, CSIR	Apr 2022 – Dec 2022
Research Assistant Professor — Marquette University	Mar 2018 – Dec 2021
Postdoctoral Fellow — Marquette University (Advisor: Dr. Murray Blackmore)	Sep 2014 – Mar 2018

EDUCATION

Ph.D., Molecular Neuroscience — University of Wisconsin, Milwaukee Thesis: Gene regulatory pathways driving CNS regeneration in Zebrafish. Advisor: Dr. Ava J. Udvadia	Aug 2009 – Sep 2014
B.Tech, Biotechnology — Anna University	May 2005 – May 2009

PREPRINTS

* indicates equal contribution · * indicates Ishwariya as corresponding author

- Sahu, Y.; Narayan, A.P.S.; Kumaran, M.; Soni, S.; Chermakani, P.; Kesireddy, D.K.; Konda, M.; Menon, A.S.; **Venkatesh, I.*** Combinatorial Control of Corticospinal Axon Growth by Retinoic Acid Receptors. *bioRxiv* 2026.07.24.740479, doi: [10.64898/2026.07.24.740479](https://doi.org/10.64898/2026.07.24.740479) (In review, PNAS)
- Menon, A.S.; Kumaran, M.; Farooq, F.; Beji, D.S.; Kesireddy, D.K.; Sahu, Y.; **Venkatesh, I.** Rewiring of the three-dimensional genome encodes regenerative potential in the adult central nervous system. *bioRxiv* doi: [10.64898/2026.03.12.711224](https://doi.org/10.64898/2026.03.12.711224) (In revision, Nature Communications)
- Krishna, N.*; Menon, A.S.*; Kumaran, M.; Kesireddy, D.; **Venkatesh, I.** PATZ1 remodels the nucleosome landscape to promote chromatin accessibility in injured neurons. *bioRxiv* 2025.12.22.695949, doi: [10.64898/2025.12.22.695949](https://doi.org/10.64898/2025.12.22.695949) (In revision, Communications Biology)
- Sanyal, K.+; Haque, S.R.+; Pendharkar, R.; Chermakani, P.; Kumaran, M.; Acharya, P.; Maity, D.*; **Venkatesh, I.*** Targeting RAR β With a Stable Retinoic Acid Mimic Promotes Neurite Outgrowth, Neurogenesis, and Behavioral Recovery. *bioRxiv* 2025.11.05.686894, doi: [10.1101/2025.11.05.686894](https://doi.org/10.1101/2025.11.05.686894) (In revision, ACS Chem Neuro)
- Kumaran, M.; Narayan, A.P.S.; Sahu, Y.; Banerjee, S.; Menon, A.S.; Soni, S.; **Venkatesh, I.*** Deep Neurite Analysis Tool (DeNAT): A machine-learning framework for precise automated neurite outgrowth measurement. *bioRxiv* doi: [10.1101/2025.09.18.676567](https://doi.org/10.1101/2025.09.18.676567) (In revision, BMC Bioinformatics)
- Menon, A.S.; Kumaran, M.+; Beji, D.S.+; Kesireddy, D.+; Sahu, Y.; Manjunath, S.; Sanyal, K.; Konda, M.; Banerjee, S.; **Venkatesh, I.*** PATZ1 Reinstates a Growth-Permissive Chromatin Landscape in Adult Corticospinal Neurons After Injury. *bioRxiv* 2025.05.14.654166, doi: [10.1101/2025.05.14.654166](https://doi.org/10.1101/2025.05.14.654166) (In revision, BMC Genome Biology)
- Sahu, Y.; Banerjee, S.+; Kumaran, M.+; Sanyal, K.; Konda, M.; Soni, S.; Menon, A.S.; Manjunath, S.; Kesireddy, D.K.; Beji, D.S.; Chermakani, P.; Biswas, A.; **Venkatesh, I.*** Nuclear Receptor Transcription factors promote axon regeneration in the Adult Corticospinal Tract. *bioRxiv* 2025.06.19.660319, doi: [10.1101/2025.06.19.660319](https://doi.org/10.1101/2025.06.19.660319) (In revision, EMBO J)

PUBLICATIONS

- Wang, Z.; Kumaran, M.; Batsel, E.; Testor-Cabrera, S.; Beine, Z.; Alvarez Ribelles, A.; Tsoulfas, P.; **Venkatesh, I.***; Blackmore, M.G.* Single-Nuclei Sequencing Reveals a Robust Corticospinal Response to Nearby Axotomy but Overall Insensitivity to Spinal Injury. *J. Neurosci.* 2025, e1508242024. [10.1523/JNEUROSCI.1508-24.2024](https://doi.org/10.1523/JNEUROSCI.1508-24.2024)
- Venkatesh, I.***; Mehra, V.; Wang, Z.; Simpson, M.T.; Eastwood, E.; Chakraborty, A.; Beine, Z.; Gross, D.; Cabahug, M.; Olson, G.; Blackmore, M.G.* Co-Occupancy Identifies Transcription Factor Co-Operation for Axon Growth. *Nat Commun* 2021, 12(1), 2555. [10.1038/s41467-021-22828-3](https://doi.org/10.1038/s41467-021-22828-3)
- Venkatesh, I.***; Mehra, V.; Wang, Z.; Califf, B.; Blackmore, M.G. Developmental Chromatin Restriction of Pro-Growth Gene Networks Acts as an Epigenetic Barrier to Axon Regeneration in Cortical Neurons. *Dev Neurobiol* 2018, 78(10), 960–977. [10.1002/dneu.22605](https://doi.org/10.1002/dneu.22605)
- Wang, Z.; Mehra, V.; Simpson, M.T.; Maunze, B.; Chakraborty, A.; Holan, L.; Eastwood, E.; Blackmore, M.G.*; **Venkatesh, I.*** KLF6 and STAT3 Co-Occupy Regulatory DNA and Functionally Synergize to Promote Axon Growth in CNS Neurons. *Sci Rep* 2018, 8(1), 12565. [10.1038/s41598-018-31101-5](https://doi.org/10.1038/s41598-018-31101-5)
- Venkatesh, I.***; Blackmore, M.G. Selecting Optimal Combinations of Transcription Factors to Promote Axon Regeneration: Why Mechanisms Matter. *Neurosci Lett* 2017, 652, 64–73. [10.1016/j.neulet.2016.12.032](https://doi.org/10.1016/j.neulet.2016.12.032)
- Venkatesh, I.***; Simpson, M.T.; Coley, D.M.; Blackmore, M.G. Epigenetic Profiling Reveals a Developmental Decrease in Promoter Accessibility during Cortical Maturation in Vivo. *Neuroepigenetics* 2016, 8, 19–26. [10.1016/j.nepig.2016.10.002](https://doi.org/10.1016/j.nepig.2016.10.002)
- Simpson, M.T.+; **Venkatesh, I.***; Callif, B.L.; Thiel, L.K.; Coley, D.M.; Winsor, K.N.; Wang, Z.; Kramer, A.A.; Lerch, J.K.; Blackmore, M.G. The Tumor Suppressor HHEX Inhibits Axon Growth When Prematurely Expressed in Developing Central Nervous System Neurons. *Mol Cell Neurosci* 2015, 68, 272–283. [10.1016/j.mcn.2015.08.008](https://doi.org/10.1016/j.mcn.2015.08.008)
- Williams, R.R.+; **Venkatesh, I.***; Pearse, D.D.; Udvadia, A.J.; Bunge, M.B. MASH1/Ascl1a Leads to GAP43 Expression and Axon Regeneration in the Adult CNS. *PLoS One* 2015, 10(3), e0118918. [10.1371/journal.pone.0118918](https://doi.org/10.1371/journal.pone.0118918)

9. **Venkatesh, I.†**; Makky, K. Teaching Epigenetic Regulation of Gene Expression Is Critical in 21st-Century Science Education: Key Concepts & Teaching Strategies. *The American Biology Teacher* 2020, 82(6), 372–380. [10.1525/abt.2020.82.6.372](https://doi.org/10.1525/abt.2020.82.6.372)

AWARDS & RECOGNITION

- **2025** — Best Paper Award from Organizers, Indian Academy of Neurosciences
- **2025** — Best Lab Exhibit on Open Day, CCMB
- **2024** — Best Paper Award from Organizers, Indian Academy of Neurosciences
- **2024** — Best Lab Exhibit on Open Day, CCMB
- **2023** — Invited to editorial board, Journal of Biosciences, Indian Academy of Sciences
- **2023** — Elected Associate, Indian Academy of Sciences
- **2023** — Best Paper Award from Organizers, Indian Academy of Neurosciences
- **2023** — Best Lab Exhibit on Open Day, CCMB
- **2022** — Best Paper Award from Organizers, Indian Academy of Neurosciences
- **2016** — Invited review panel member, NSF Bioinformatics, National Science Foundation (USA)
- **2014** — UWM Chancellor's Award for Graduate Research, University of Wisconsin-Milwaukee
- **2013** — Best Research Poster Award, International Society for Neuroregeneration

GRANT FUNDING

Ongoing (Role: Principal Investigator)

- Department of Biotechnology (2024–2027) — Nuclear Receptors mediated molecular regulation of nervous system development and disease
- BFIbome (2026–2027) — Bioactive Scaffolds combined with Gene Therapy for Spinal Cord Repair
- ANRF – Advanced Research Grant (2026–2029)
- Department of Biotechnology (2026–2029)

Approved to Be Sanctioned

- CSIR – Focused Basic Research Grant (FBR), 2026–2028

Completed as PI, India

- CSIR Neuromission (2024–2026)
- SERB SRG (2024–2026)
- CSIR OLP (2022–2025)
- India Bioscience Outreach Grant (2022–2023)
- SERB Ramanujan Fellowship (Apr 2022–Dec 2022)

Completed, United States

- NSF – XSEDE XRAC grant (2020–2022)
- NIH R21, Co-Investigator — R21NS106309 (2018–2020)
- NSF – XSEDE XRAC grant (2018–2020)
- Craig H. Nielsen Postdoctoral Fellowship (2016–2018)
- NSF – XSEDE grant for Bioinformatics Data Analysis (2016–2018)
- Ruth Walker Grant-in-Aid for Graduate Research (2012–2014)
- UWM Chancellor's Award for Graduate Research (2009–2014)

ACADEMIC SERVICES

Editorial & Review

- Editorial Services: The Journal of Biosciences
- Manuscript Reviews: The Journal of Biosciences, Nature Communications, Communications Biology
- Thesis Reviews: Tata Institute for Fundamental Research (TIFR), Mumbai; Aravind Eye Medical Research Foundation (AMRF), Madurai
- Grant Reviews: Department of Biotechnology (DBT), Biomedical Sciences cohort; Council of Scientific and Industrial Research (CSIR); Expert Committee Member, Life Sciences (ANRF – NPDP)
- CSIR Taskforce Grant Review Committee Member — Translational Neuroscience: Pharmacological Validation of Key Biological Pathways Implicated in Neurodegenerative Diseases

Memberships

- Society for Neuroscience (2009–2021)
- Indian Academy of Neurosciences (2022–Life)
- Society for Neurochemistry (2025–Life)

Doctoral Thesis Committee Member

- Ashwani Kumar (Dr. Santosh Chauhan group, CSIR-CCMB)
- Bithan Rakshit (Dr. Vinay Kumar Nandicoori group, CSIR-CCMB)
- Shantanu Vishal (Dr. Janesh Kumar group, CSIR-CCMB)
- Debatri Dutta (Dr. Vinay Kumar Nandicoori group, CSIR-CCMB)

- Yogita Sharma (Dr. Rajan Sankaranarayan group, CSIR-CCMB)
- Sukriti Agarwal (Dr. Anindhya Ghosh Roy group, NBRC-Gurgaon)

Judging Duties

- Oral Presentation Judge, HySci 2024, IIT-Hyderabad

Committee Services at CSIR-CCMB

- Tissue Culture Facility Committee, Member (2025–Present)
- Local Purchase Committee, Member (2024–Present)
- Microscopy Committee, Member (2024–Present)
- Animal House Committee, Member (2024–Present)
- Scientific Ethics and Vigilance Committee, Convener (2024–Present)
- Management Council, CSIR-CCMB, Member (2025–Present)
- Institutional Project Associate Recruitment Committees (2024, 2025)
- Institutional Ph.D. Student Interview Committee (2022–Present)
- Institutional Recruitment Committee, Group III (2025)

SCIENCE COMMUNICATION & OUTREACH

2025

- CCMB Open Day — Exhibit "Know Your Brain" (1st Prize)
- Hosted students, Young Innovator's Program, CCMB
- Shadow a Scientist Program — hosted a high school student (March 2025)
- "Why Every Life Science Trainee Needs a Personal Website" — article for SciTales, CCMB ([link](#))
- "From Concept to Creation: Lab Members Build Personal Websites" — article for SciTales, CCMB ([link](#))

2024

- Expert Mentor in Neuroscience, PhD Cafe Seminar Series, India Bioscience ([link](#))
- DoScience, Hyderabad — talk and exhibit, "Know Your Brain" ([link](#))
- CCMB Open Day — Exhibit "Know Your Brain" (2nd Prize; featured in news, [link](#))
- "How to Manage a Research Lab as a PI in India?" — article for IndiaBioscience ([link](#))

2023

- Exhibit "Know Your Brain," CSIR One Week One Lab (OWOL), CCMB
- CCMB Open Day — Exhibit "Know Your Brain" (2nd Prize)

2022

- CCMB Open Day — Exhibit "Know Your Brain"

Institutional Services

- 2025 — Represented CCMB at CSIR Women's Day events, nominated by the Director
- 2023 — Represented CCMB at CSIR Women's Day events, nominated by the Director

Open-Source Resources

- Co-authored a faculty handbook helping new CCMB faculty navigate lab setup, procurement, hiring, teaching, and campus life
- Compiled and shared a resource hub covering data visualization, scientific writing, research productivity, funding opportunities, and mentorship tools

Press

- The Hindu, 2021 — "How to recover from a spinal cord injury," featuring "Co-occupancy identifies transcription factor co-operation for axon growth"

Workshops Organized

- Poster Design Virtual Workshop, sponsored by IndiaBioscience — "Poster Design 101" (Feb 2024, [recording](#))

Panelist

- HySci 2024 — "Three Wise Men: review papers, experimental papers, and impact factors. How do they matter in the current world?"

TEACHING

At CCMB

- 2023 — Gene Regulation (Module: 3D Organization of the Genome)
- 2024 — Cell Biology (Module: Cytoskeleton and Cell Culture)
- 2025 — Cell Biology (Module: Cytoskeleton and Cell Culture)

Before CCMB

- Lead Teaching Assistant, Human Anatomy & Physiology, University of Wisconsin-Milwaukee (Fall 2010–Fall 2014) — managed 15–20 TAs for a course of 600–700 students/semester; prepared lab materials, maintained the online TA resource hub, mentored incoming TAs, ran weekly meetings, and coordinated with the senior lecturer and lab manager

INVITED TALKS

2023

- Pasteur Institute Meeting, CCMB
- Indian Academy of Neurosciences, Gwalior, MP
- Aravind Medical Research Foundation, Madurai, TN
- Indian Academy of Sciences (IASc)
- India Investigators Network (IIN) Symposium

2024

- Cellular and Molecular Mechanisms of Development and Regeneration Conference, Shiv Nadar University, Delhi NCR
- Society for Developmental Biologists, inSTEM, Bangalore
- bioAnveshna 2024 Plenary Talk, University of Hyderabad
- HySci — Hyderabad Science Meeting, IIT-Hyderabad
- VigyanShaala — Reimagining Possibilities for Women in STEM Careers, Osmania University, Hyderabad
- International Society of Regenerative Biology (Virtual Seminar Series)
- Indian Academy of Neurosciences Meeting (IAN), NIMHANS, Bangalore
- Indian Academy of Sciences, Editorial Board Talks, NISER Bhubaneswar
- International Conference on Intra and Inter-Cellular Regulatory Systems (IICRS), University of Hyderabad
- Brain and Society: Clinical and Physiological Perspective of Brain Disorders, National Brain Research Center (NBRC), Gurgaon

2025

- SYNOPSIS Lecture Series, BITS Pilani, Hyderabad Campus — March
- Departmental Seminar Series (Biological Sciences), IIT-Hyderabad — March
- National Seminar on Epigenetic Landscapes, Utkal University, Bhubaneswar — March
- Bannari Amman Institute of Technology, International Conference on Biotechnology — Keynote Speaker and Chief Guest, April
- PSG College of Technology, Coimbatore — Seminar Speaker, April
- CCMB Microscopy Course — September

2026

- Cellular and Molecular Mechanisms of Development and Regeneration Conference, Shiv Nadar University, Delhi NCR
- Synapse, IIT Tirupati
- International Conference on Translational Neuroscience (ICTN 2026): Brain to Behaviour
- BioSpan 2026, IIT-Bhilai

MENTORING

Ph.D. Students, CSIR-CCMB

- **Yogesh Sahu** (2022–present) — Best Paper Award, Indian Academy of Neurosciences (2022); Best Paper Award, Society for Neurochemistry (SNCI) Hyderabad Chapter (2025); Short Talk Travel Award, Indian Academy of Neurosciences (2025); Invited Speaker, NRSM ACTREC-Mumbai (2025)
- **Anisha Menon** (2023–present) — Best Paper Award, Indian Academy of Neurosciences (2023); Best Paper Award, CMMDR, Shiv Nadar (2023); Best Poster Award, CMMDR (2026)
- **Rutuja Pendharkar** (2024–present) — Travel Award, Indian Academy of Neurosciences (2024); Best Paper Award, Society for Neurochemistry (SNCI) Hyderabad Chapter (2025); Best Paper Award, Indian Academy of Neurosciences (2025)
- **Aarthi Sukumar** (2025–present)

Postdoctoral Fellows

- **Manojkumar Kumaran** (2022–present) — Travel Award, Indian Academy of Neurosciences (2024); published in Journal of Neuroscience (2025); Best Poster Award, CMMDR (2026); Lalitha and Ravindranath International Travel Award (2026)
- **Shringika Soni** (2024–present) — SERB National Postdoctoral Fellowship (2024); She Inspires Award, Rukhmabai Initiatives–IndiaBioscience (2024); John Miller Travel Award, Indian Academy of Neurosciences (2025)
- **Prakash C** (2024–present) — CSIR-RA Fellowship (2025–2027)
- **Praveena Koyyada** (2025–present)

Dissertation / Project Training Fellows

- Meera Manoj
- Achuth Krishna

Project Associates

- Dhruvakumar Kesireddy
- Faheem Farooq
- Athul Narayanan (VIT-Vellore) — Best Poster Award, CMMDR (2026)

Former Lab Members

- Sanjana Sinha (late)
- Sneha Manjunath — Thermo Fisher Scientific
- Katha Sanyal — Project Associate, CCMB
- Shaik Shafiulla — Ph.D. student, Germany
- Ishan Dutta — M.Tech, IIT-Warangal
- Sanskruti Karwa — Research Officer, Gennova Biopharmaceuticals
- Aastha Khiwani — Ph.D. student, PGIMER Chandigarh
- Arupam Biswas — Ph.D. candidate, Johannes Gutenberg University Mainz, Germany
- Soupayan Banerjee — Ph.D. student, Texas A&M
- Deepta Beji — Masters in International Medicine, EU
- Meghana Madhu — Ph.D. student, Umeå University, Sweden
- Pratikhya Acharya — Ph.D. student, Aix-Marseille University, France
- Netra Krishna (2026) — Ph.D. student, University of Göttingen, International Max Planck Research School, Neurosciences

Summer Trainees

- Pranjali Dubey (University of Delhi)
- Nandana Raghunandan (University of Kerala)
- Riya Kushawa (University of Delhi)
- Netra Krishna (KREA University)
- Apoorva Atmuri (M.Tech, IIT Hyderabad)
- Katha Sanyal (Project Associate, CCMB)
- Abhilaya Makkkuva (Center for Human Genetics, Bangalore)
- Tunu Vinit Reddy